Theoretical Questions

I. Mechanics

1. What are conservative forces? (Lecture on 2026-02-16, slide 6)
2. What are dissipative forces? (Lecture on 2026-02-16, slide 6)
3. What is the concept of a rigid body? (Lecture on 2026-02-16, slide 6)
4. What is the centre of mass? (Lecture on 2026-02-16, slide 6)
5. What is angular momentum? (Lecture on 2026-02-16, slide 6)
6. What is a free particle? (Lecture on 2026-02-16, slide 15)
7. What quantities describe a free particle? (Lecture on 2026-02-16, slide 15)
8. What is kinetic energy? (Lecture on 2026-02-16, slide 15)
9. What is momentum? (Lecture on 2026-02-16, slide 15)
10. What is the dispersion relation conceptually? (Lecture on 2026-02-16, slide 16)
11. What does Newton's first law state? (Lecture on 2026-02-16, slide 17)
12. What is inertia? (Lecture on 2026-02-16, slide 17)
13. What does Newton's second law describe? (Lecture on 2026-02-16, slide 17)
14. What is the meaning of force in Newtonian mechanics? (Lecture on 2026-02-16, slide 17)
15. What is Newton's third law? (Lecture on 2026-02-16, slide 17)
16. What is conservation of momentum? (Lecture on 2026-02-16, slide 18)
17. What is mechanical work? (Lecture on 2026-02-16, slide 18)
18. What is the work-energy principle? (Lecture on 2026-02-16, slide 19)
19. How does work affect kinetic energy? (Lecture on 2026-02-16, slide 19)
20. What is mechanical energy? (Lecture on 2026-02-16, slide 20)
21. When is mechanical energy conserved? (Lecture on 2026-02-16, slide 21)
22. What is gravity as an interaction? (Lecture on 2026-02-16, slide 22)
23. What did Kepler discover about planetary motion? (Lecture on 2026-02-16, slide 22)
24. What was Newton's contribution to gravity? (Lecture on 2026-02-16, slide 23)
25. What are conic sections in orbital motion? (Lecture on 2026-02-16, slide 25)
26. What did Galileo demonstrate about free fall? (Lecture on 2026-02-16, slide 27)
27. Why do objects fall with the same acceleration? (Lecture on 2026-02-16, slide 27)
28. What is Newton's law of gravitation conceptually? (Lecture on 2026-02-16, slide 28)
29. What is gravitational potential energy? (Lecture on 2026-02-16, slide 30)
30. Why is gravity a conservative force? (Lecture on 2026-02-16, slide 30)
31. What is escape velocity? (Lecture on 2026-02-16, slide 31)
32. What does escape velocity depend on? (Lecture on 2026-02-16, slide 31)
33. What is a dissipative force field? (Lecture on 2026-02-16, slide 32)

34. Give examples of dissipative forces. (Lecture on 2026-02-16, slide 33)
35. Why do dissipative forces reduce energy? (Lecture on 2026-02-16, slide 32)
36. What is mechanical power? (Lecture on 2026-02-16, slide 34)
37. What is a system of point masses? (Lecture on 2026-02-23, slide 14)
38. What are internal and external forces? (Lecture on 2026-02-23, slide 15)
39. What is a rigid body? (Lecture on 2026-02-23, slide 16)
40. What is torque? (Lecture on 2026-02-23, slide 17)
41. What is rotational motion? (Lecture on 2026-02-23, slide 17)
42. What is angular momentum of a rigid body? (Lecture on 2026-02-23, slide 18)
43. What is the parallel axis theorem? (Lecture on 2026-02-23, slide 18)
44. What is rotational kinetic energy? (Lecture on 2026-02-23, slide 19)
45. What is equilibrium of a rigid body? (Lecture on 2026-02-23, slide 17)
46. What is a harmonic oscillator? (Lecture on 2026-02-23, slide 21)
47. What is the restoring force? (Lecture on 2026-02-23, slide 21)
48. What is angular frequency? (Lecture on 2026-02-23, slide 22)
49. What is damping? (Lecture on 2026-02-23, slide 27)
50. What causes damping in real systems? (Lecture on 2026-02-23, slide 26)
51. What is resonance? (Lecture on 2026-02-23, slide 35)
52. Why does resonance increase amplitude? (Lecture on 2026-02-23, slide 35)
53. Give real-life examples of resonance. (Lecture on 2026-02-23, slide 37)
54. What is Fourier decomposition conceptually? (Lecture on 2026-02-23, slide 40)
55. Why can periodic signals be decomposed into sinusoids? (Lecture on 2026-02-23, slide 40)

II. Thermodynamics

1. What are thermodynamic state variables? (2026-03-09, slide 20)
2. What is temperature physically? (2026-03-09, slide 20)
3. What is the Celsius scale? (2026-03-09, slide 1)
4. What is the Kelvin scale? (2026-03-09, slide 3)
5. What is thermal expansion? (2026-03-09, slide 5)
6. What is linear expansion? (2026-03-09, slide 5)
7. What is volumetric expansion? (2026-03-09, slide 7)
8. What is a thermometer? (2026-03-09, slide 8)
9. What is a resistance thermometer? (2026-03-09, slide 9)
10. What is the ideal gas model? (2026-03-09, slide 20)
11. What assumptions define an ideal gas? (2026-03-09, slide 20)
12. What is the ideal gas equation conceptually? (2026-03-09, slide 21)
13. What is an isothermal process? (2026-03-09, slide 22)
14. What is an isobaric process? (2026-03-09, slide 23)

15. What is an isochoric process? (2026-03-09, slide 24)
16. What is specific heat? (2026-03-09, slide 26)
17. What is heat capacity? (2026-03-09, slide 27)
18. What is internal energy? (2026-03-16, slide 10)
19. What does the first law of thermodynamics state? (2026-03-16, slide 11)
20. What is enthalpy? (2026-03-16, slide 15)
21. Why is enthalpy useful? (2026-03-16, slide 15)
22. What is an adiabatic process? (2026-03-16, slide 18)
23. What happens to temperature during adiabatic expansion? (2026-03-16, slide 18)
24. What is a thermodynamic cycle? (2026-03-23, slide 7)
25. What is thermal efficiency? (2026-03-23, slide 8)
26. What is the Carnot cycle? (2026-03-23, slide 9)
27. Why is the Carnot cycle important? (2026-03-23, slide 9)
28. What is the second law of thermodynamics? (2026-03-23, slide 11)
29. Why is a perpetual motion machine impossible? (2026-03-23, slide 11)
30. What is entropy? (2026-03-23, slide 15)
31. What does entropy measure physically? (2026-03-23, slide 15)
32. What is free energy? (2026-03-23, slide 17)
33. What is kinetic gas theory? (2026-03-30, slide 7)
34. What is pressure in kinetic theory? (2026-03-30, slide 7)
35. What is Boltzmann constant? (2026-03-30, slide 8)
36. What is equipartition theorem? (2026-03-30, slide 10)
37. What is internal energy in kinetic theory? (2026-03-30, slide 11)
38. What is Boltzmann's entropy definition? (2026-03-30, slide 13)

III. Electrostatics, Waves, and Optics

1. What is electrostatic potential? (2026-02-16, slide 9)
2. What is electric charge? (2026-04-13, slide 6)
3. What are positive and negative charges? (2026-04-13, slide 6)
4. What is charge quantization? (2026-04-13, slide 7)
5. What is Coulomb's law? (2026-04-13, slide 13)
6. What determines electrostatic force? (2026-04-13, slide 13)
7. What is electric field? (2026-04-13, slide 15)
8. How is electric field defined? (2026-04-13, slide 15)
9. What are electric field lines? (2026-04-13, slide 16)
10. What does their density represent? (2026-04-13, slide 16)
11. What is Gauss's law? (2026-04-13, slide 18)
12. What is electric flux? (2026-04-13, slide 18)

13. What is a wave? (2026-03-02, slide 10)
14. What is amplitude? (2026-03-02, slide 11)
15. What is wavelength? (2026-03-02, slide 11)
16. What is frequency? (2026-03-02, slide 12)
17. What is phase velocity? (2026-03-02, slide 13)
18. What is the relation between speed, wavelength and frequency? (2026-03-02, slide 14)
19. What is reflection of waves? (2026-03-02, slide 19)
20. What is refraction? (2026-03-02, slide 20)
21. What is diffraction? (2026-03-02, slide 22)
22. What is interference? (2026-03-02, slide 23)
23. What are constructive and destructive interference? (2026-03-02, slide 23)
24. What is polarization? (2026-03-02, slide 26)
25. What is sound? (2026-03-02, slide 30)
26. Why is sound a longitudinal wave? (2026-03-02, slide 30)
27. What is the Doppler effect? (2026-03-02, slide 32)
28. What is sound intensity? (2026-03-02, slide 33)
29. What is the decibel scale? (2026-03-02, slide 34)
30. What does Weber–Fechner law describe? (2026-03-02, slide 34)